Effect of dietary patterns on measures of lipid peroxidation: results from a randomized clinical trial.

BACKGROUND: Free radical-mediated oxidative damage to lipids is thought to be an important process in the pathogenesis of atherosclerosis. Although previous studies have demonstrated a beneficial impact of antioxidant vitamin supplements on lipid peroxidation, the effect of dietary patterns on lipid peroxidation is unknown. METHODS AND RESULTS: During the 3-week run-in period of a randomized trial, 123 healthy individuals were fed a control diet, low in fruits, vegetables, and dairy products, with 37% of calories from fat. Participants were then randomized to consume for 8 weeks: (1) the control diet, (2) a diet rich in fruits and vegetables but otherwise similar to the control diet, and (3) a combination diet rich in fruits, vegetables, and low-fat dairy products and reduced in fat. Serum oxygen radical-absorbing capacity, malondialdehyde (an in vitro measure of lipid peroxidation), and breath ethane (an in vivo measure of lipid peroxidation) were measured at the end of run-in and intervention periods. Between run-in and intervention, mean (95% CI) change in oxygen radical-absorbing capacity (U/mL) was -35 (-93, 13) in the control diet, 26 (-15, 67) in the fruits and vegetables diet (P=0.06 compared with control), and 19 (-22, 54) in the combination diet (P=0.10 compared with control). Median (interquartile range) change in ethane was 0.84 (0.10, 1.59) in the control diet, 0.02 (-0.61, 0.83) in the fruits and vegetables diet (P=0.04 compared with control), and -1.00 (-1.97, 0.25) in the combination diet (P=0.005 compared with control). Change in malondialdehyde did not differ between diets. CONCLUSIONS: This study demonstrates that modification of diet can favorably affect serum antioxidant capacity and protect against lipid peroxidation.